CONCEPTION, DESIGN AND MANUFACTURE OF TYPE IV HYDROGEN PRESSURE VESSELS FOR GENERAL AVIATION

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Abstract

Hydrogen plays an important role in decarbonising general aviation but demands more storage space compared to state of the art jet fuel. Geometrically optimised cylindrical or ellipsoidal vessel designs show potential solutions for the challenge of maximum utilisation of installation space and pose new scientific questions regarding design, design methodology and the manufacturing process to achieve high specific storage densities of the vessel system. The design process of such optimised cylindrical vessel is introduced under consideration of important regulatory requirements. Material and manufacturing process choices for the designed type IV vessel in regards to liner and carbon fibre reinforced polymer (CFRP)-shell are discussed and applied.

1. INTRODUCTION

According to recent studies, hydrogen will have an important role to reduce climate footprint of aviation both on the ground and for flights as well [1]. To enable the use of gaseous hydrogen as fuel for flight missions, it is necessary to store the hydrogen as efficient as possible. One option to increase storage density is to store hydrogen within pressurised vessels. To achieve an improved lightweight design those vessels can be made of fibre-reinforced plastics (FRP) as those materials combine high specific properties such as tensile strength and stiffness with low density and therefore promise high weight saving potential.

Due to the fact, that the possible weight saving potential comes with a more difficult conception, design, and manufacturing phase, this paper shall give an overview on the methodology on how to develop high performance hydrogen vessels under a given design space for a general aviation application.

1.1. Typical structure of type IV hydrogen vessels

As displayed in FIGURE 1 and in accordance to Fowler et al [2] pressurised hydrogen vessel can be classified in five groups with increasing proportion of FRP-laminate from type I to V.

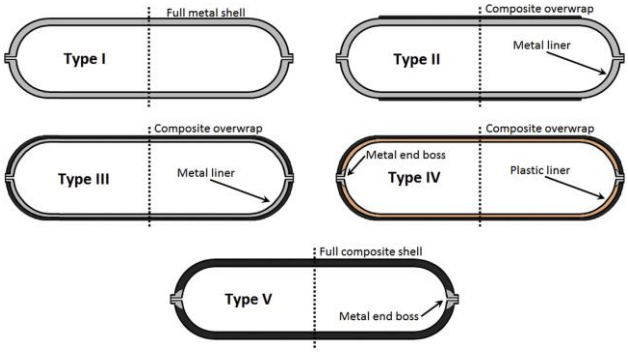

FIGURE 1. Classification of hydrogen vessels into type I -V [2]

While type V vessels are still an academic approach without wide application range due to challenges with leakage and permeation properties within the FRP-laminate under cyclic stress, type IV vessels are already used in automotive applications [3]. The better feasibility of this approach lies within the function separation between the usually carbon fibre reinforced plastic (CFRP) -hull, which is responsible for mechanical strength, whereas the compared to type V vessels added polymer liner is responsible for hydrogen leakage respectively permeation tightness. To enable the connection to valves and piping system metallic mounting elements, so called boss parts complete the in FIGURE 2 displayed typical structure of type IV hydrogen vessels.

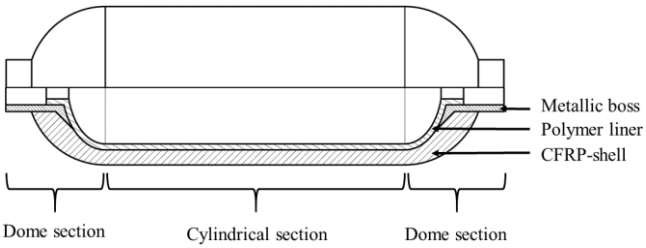

FIGURE 2. Typical structure of type IV hydrogen vessels [4]

1.2. Definition and classification of target figures

Besides stored amount of hydrogen, cost, operational time and environmental impact there are two other metrics that allow the comparison of hydrogen vessels. The first one is the gravimetric storage density η_{gravim}. This metric describes how much hydrogen can be stored within a vessel in relation to the weight of the storage system including piping and valves. The gravimetric storage density of a hydrogen vessel can be calculated by:

(2) $\eta_{gravim} = \frac{m_{H_2}}{m_{system}}$

The second metric is the volumetric storage utilisation η_{volum} that determines the usage of a given build space by the hydrogen storage system. The volumetric storage utilisation is given by:

(3) $\eta_{volum} = \frac{V_{stored\,H_2}}{V_{build\,space}}$

Those two metrics need to be considered in context with each other to achieve an optimum of stored amount of hydrogen and weight of the storage system depending on the specific mission and design problem.

Especially for class CS23 general aviation aircrafts like the Antares E2 manufactured by Lange Research Aircraft given build space for hydrogen storage is low. Therefore the volumetric storage utilisation of the available build space needs to be optimised in order to store enough fuel to achieve sufficient flight range. For this reason vessel shapes need to be considered that maximise the amount of stored hydrogen by increasing the storage utilisation. The for the mechanical properties of the CFRP-laminate not optimised vessel shape results in principle lower gravimetric storage densities of those vessel shapes.
In conclusion both gravimetric storage density and volumetric storage utilisation need to be optimised to enable the use of hydrogen for general aviation.

2. METHODS

The iterative design process of hydrogen vessels used in this paper is displayed within the flow chart in FIGURE 3. Important design choices will be detailed in the following subsections.

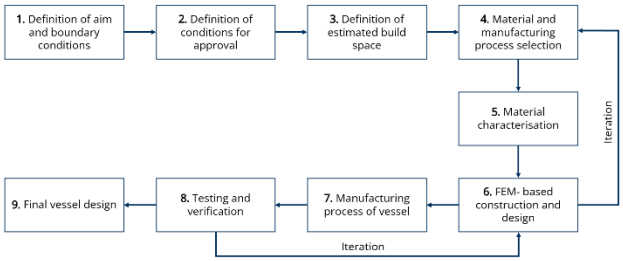

FIGURE 3. Flow chart of iterative design process of discussed hydrogen vessel

2.1. Definition of boundary conditions and safety factors for hydrogen vessels in general aviation applications

Current aircraft regulations do not provide specific rules for the testing and certification procedure of hydrogen vessels in aviation applications. For this reason UN/ECE R134 [5] is applied as basis reference for the development of such vessels as this European regulation already finds broad application within the automotive sector.

According to UN/ECE R 134 besides cyclic tests the hydrogen vessel has to withstand a minimal burst pressure BP_{min} given by:

(4) $BP_{min} = 2.25 * NWP$ [5]

where NWP is defined as the nominal working pressure of the vessel. This boundary condition has an important role for the dimensioning of the vessel besides permitted permeation properties, cyclic test regimes and further impact test demanded by UN/ECE R 134 [5].
As operating temperature of aircrafts differ significantly from operation temperature of automotive applications and have an important impact on material selection, those requirements are adapted from CS 23 [6].

2.2. Definition of an estimated build space

As stated above, the aircraft Antares E2 has restricted build space for additional fuel vessels. While the wings and body section of the aircraft are comparatively small, they do not offer sufficient space for a state of the art hydrogen pressure vessel. For this reason it is necessary to integrate the hydrogen storage in the provided outer nacelles. For aerodynamic reasons these containments need to have a specific shape. As displayed in FIGURE 4, state of the art cylindrical shaped hydrogen vessels do not achieve optimal use of the available build space. Therefore it is important especially to optimise the dimensions of the vessel and improve the build space usage. While the outer nacell will not be an integrated part of the hydrogen vessel, additional spacing distances have to be considered. To facilitate the design process these distances will be accounted from the

outer surface of the liner and consist of the following positions:

- thickness of CFRP- laminate
- design space for valve and hydrogen piping
- vessel expansion due to inner pressure
- vessel mounting system
- safety distance to hull

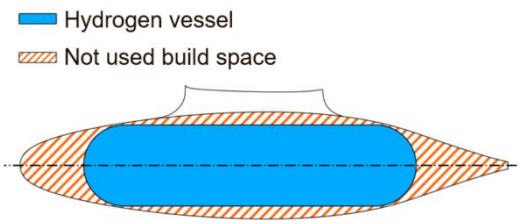

FIGURE 4. Given aerodynamic build space with dimensional optimised cylindrical hydrogen vessel

2.3. Material and manufacturing process selection

Apart from regulatory requirements the selection of materials and manufacturing process are both highly dependent on each other. Especially the manufacturing of liners shows high dependencies as not every liner material is suitable for every manufacturing process and vice versa. Furthermore material properties on component level are affected by raw material properties and by the manufacturing process as well [7]. These emerging material characteristics are the basis for the design and dimensioning process, which places further demands on the to be achieved material properties. The iterative process displayed in FIGURE 5 results in a final selection of suitable materials and manufacturing processes.

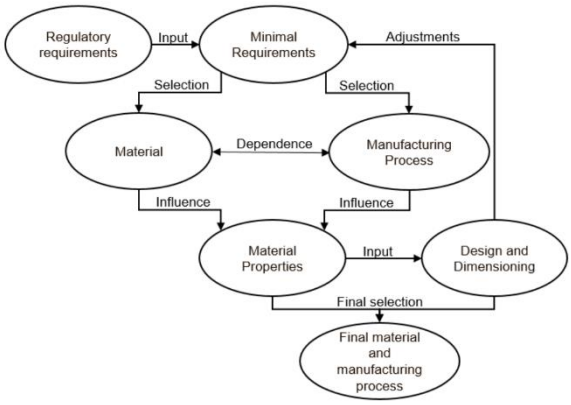

FIGURE 5. Iterative process for material and manufacturing process selection

3. RESULTS

In accordance to the outlined proceedings the design process of a cylindrical hydrogen vessel for Antares E2 was performed with corresponding material and manufacturing process selection. To increase clarity, design choices for CFRP-shell and thermoplastic liner are reviewed separately in the following explanations.

3.1.1. General design specifications

Referring to CS 23, the operating temperatures of the aircraft and therefore for the hydrogen vessel as well lie within the range of -56,5°C to 50°C [6]. These operating temperatures are extended by DIN EN 17127, which allows temperatures during refuelling process for land vehicles to up to 85°C [8].

Although DIN EN 17127 provides refuelling routine for both systems with 350 MPa and 700 MPa NWP, due to the very limited available design space 700 MPa NWP was chosen as preferred NWP. This decision is reasoned by the higher density of hydrogen under elevated pressure and the resulting possibility to store more hydrogen in total [9][9].

3.1.2. Material and manufacturing process selection of composite shell

Due to available manufacturing possibilities, good establishment in regards to the production of hydrogen vessels, low costs and high process stability, wet filament winding process is the preferred manufacturing method of the CFRP-shell.
Based on this definition additional requirements for a thermoset resin system and carbon fibres result.

The relevant requirements for suitable carbon fibres are:

- High tensile strength
- Moderate modulus for winding process
- Compatible seizing for resin system
- Sufficient yield to achieve high winding speeds
- Good availability under current carbon fibre market situation
- Moderate cost

The relevant requirements for a suitable resin system are:

- Glass transition temperature > 85°C,
- Good compatibility to carbon fibres
- Low viscosity over long time period
- Long pot life
- Elevated fracture toughness
- Good availability
- Fast curing at elevated temperatures
- Curing temperature below application limit of liner material

Evaluating several carbon fibres and resin systems specifically for this wet winding application the following material system for the CFRP-shell was chosen:

- Carbon fibres: Tenax™-E ITS50 F23 24K 1600tex D by TEIJIN CARBON EUROPE GMBH
- Resin system: Araldite® LY556 /Aradur® 917/ Accelerator DY 070 by Huntsman Advanced Materials

3.1.3. Design, material and manufacturing process selection for type IV liner

To determine suitable processes for liner production the accurate dimensions of the liner part need to be determined. According to Vasiliev [10] the thickness of the CFRP- laminate t_{lam} within the cylindrical area of a hydrogen vessel can be estimated by:

(5) $$t_{lam} = \frac{3*p*D_i}{4*\bar{\sigma}_l}$$ [10]

in which p is replaced by BP_{min} from (4), D_i is the inner diameter of the vessel and $\bar{\sigma}_l$ is the unidirectional tensile strength of the used composite in fibre direction.

The final optimised dimensions of the liner geometry is set to 410 mm in outer diameter and a total length of 1800 mm. The design and complex optimisation of the polcap-shape will be detailed within a further publication.

It can be anticipated that the iterative vessel development leads to adjustments in the detailed design of the vessel shape especially in the polcap-area. As this also affects the liner geometry an important requirement for the liner manufacturing besides cost efficiency for small lot sizes lies within a high shape adjustability for different iteration generations. This requirement cannot be fulfilled by state of the art cost intense liner production like rotational moulding or blow moulding process.

For this reason a new approach based on the 3D-printing process is evaluated and implemented with positive outcome. Detailed results will be published in a further scientific publication.

To fulfil operating temperature, mechanical and permeation requirements both PA6 and PA11 are considered as possible liner material. Due to specific additional requirements of the 3D-printing process, carbon-short-fibre-reinforced material is tested. Besides the permeation properties especially the elongation at fracture is defined as an important metric to compare both materials. The final evaluation results in PA 11 with 20% carbon fibre short fibre reinforcement as preferred material.

4 CONCLUSIONS AND OUTLOOK

A method for the development of build space optimised cylindrical type IV hydrogen vessels was introduced. Based on requirements for automotive and aviation regulations fundamental design, material and manufacturing process selections were displayed and detailed. Additional material requirements were defined in order to address manufacturing processes demands and limitations. Application specific design and material recommendations were given based on gathered information. While the verification of the presented research results for the optimised cylindrical vessel is still pending, further research in accordance to Schlegel et al [4] show high potential in the realisation of an ellipsoidal vessel that has an higher volumetric storage utilisation with predicted good gravimetric storage density as displayed in FIGURE 6.

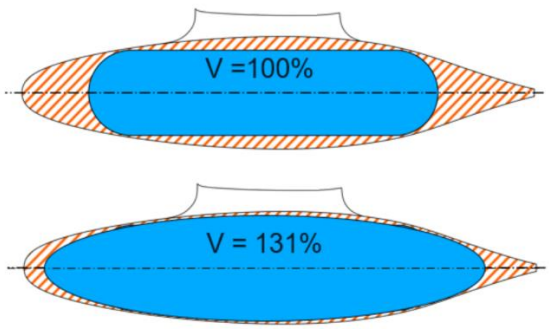

FIGURE 6. Comparison of volumetric storage utilisation of cylindric (top) and ellipsoidal (bottom) vessel geometry

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